Practical Report

WEB FUNDAMENTALS

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Scope: One combined report for the two related TryHackMe rooms under Web Fundamentals. Insert the original screenshots at the evidence points before submitting.

Room 1: Web Application Basics

1. Objective

To understand how web applications work and how browsers communicate with web servers through HTTP requests and responses.

2. Concepts and Understanding

I learned the basic structure of web applications, URL components, HTTP methods, status codes, request and response headers, and the role of security headers. I also understood how these components help explain web application behavior and identify potential security misconfigurations.

3. Practical Work

I completed practical exercises involving HTTP requests, response headers, and different request methods. I also reviewed my university website's security-header configuration using SecurityHeaders.com to understand how security headers are checked and reported.

4. Tools and Resources Used

- Tryhackme.com
- Web browser
- SecurityHeaders.com

5. Evidence and Screenshot

There are some relevant screenshot and evidence.

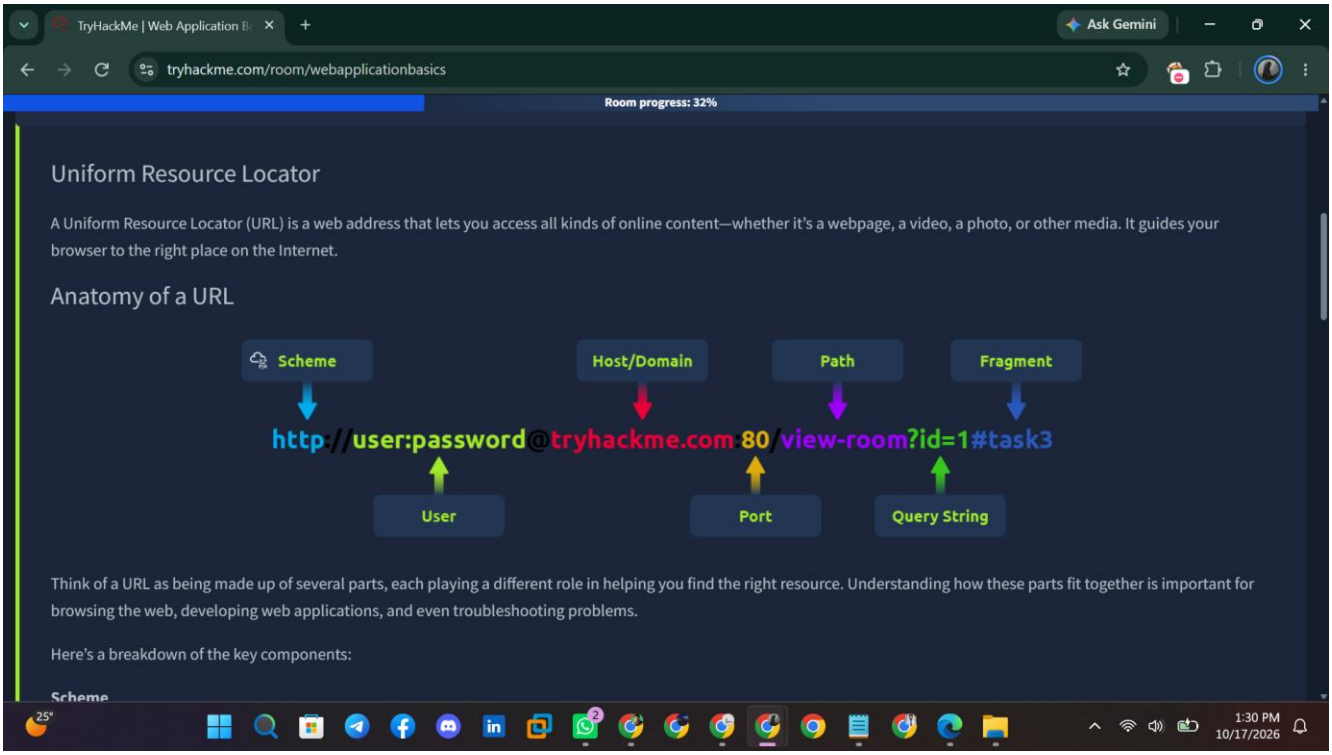


Figure 1: URL anatomy

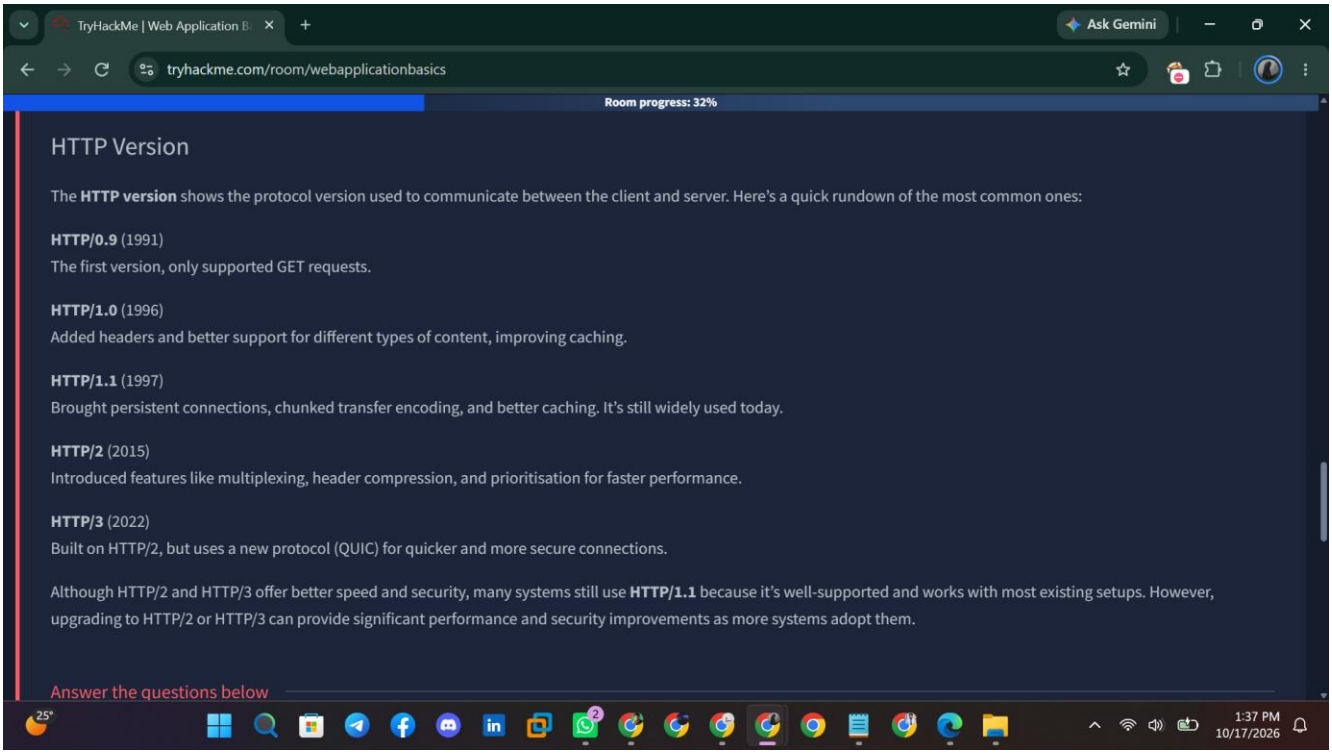


Figure 2: HTTP versions

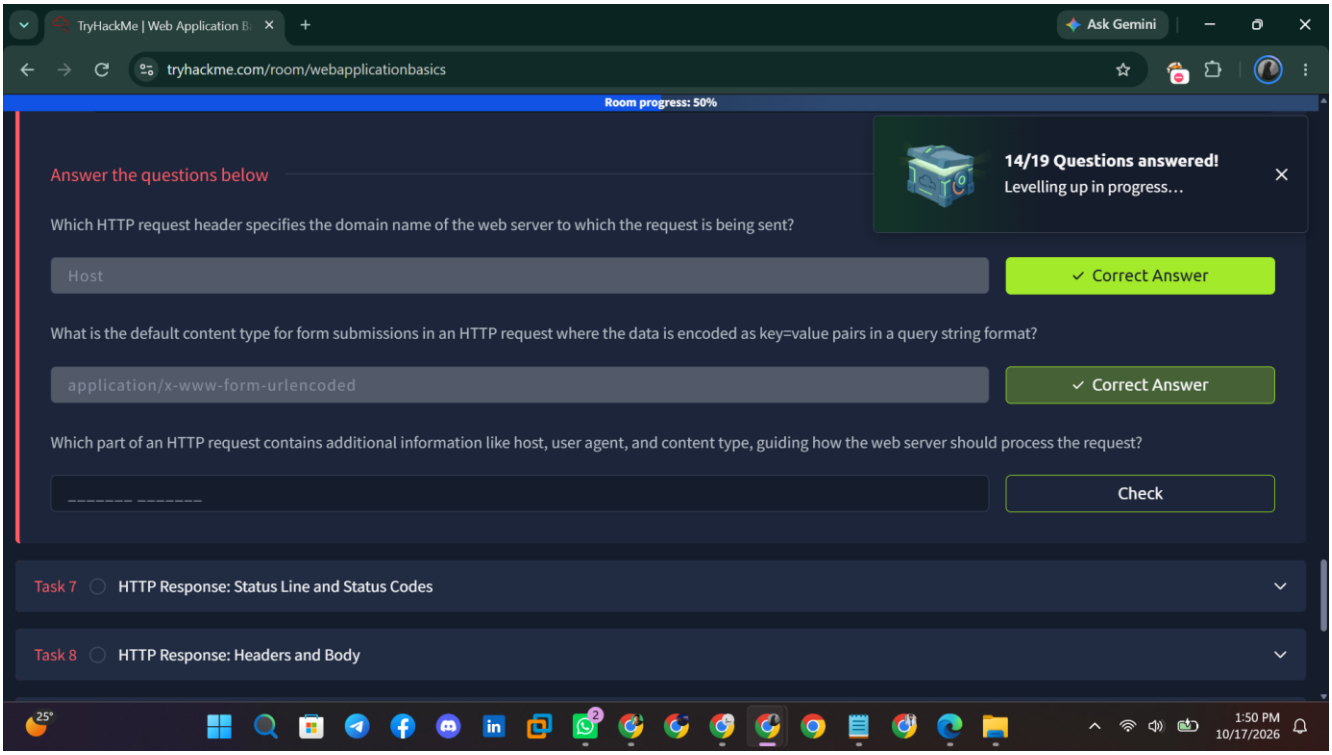


Figure 3: Http headers and body explore and understand

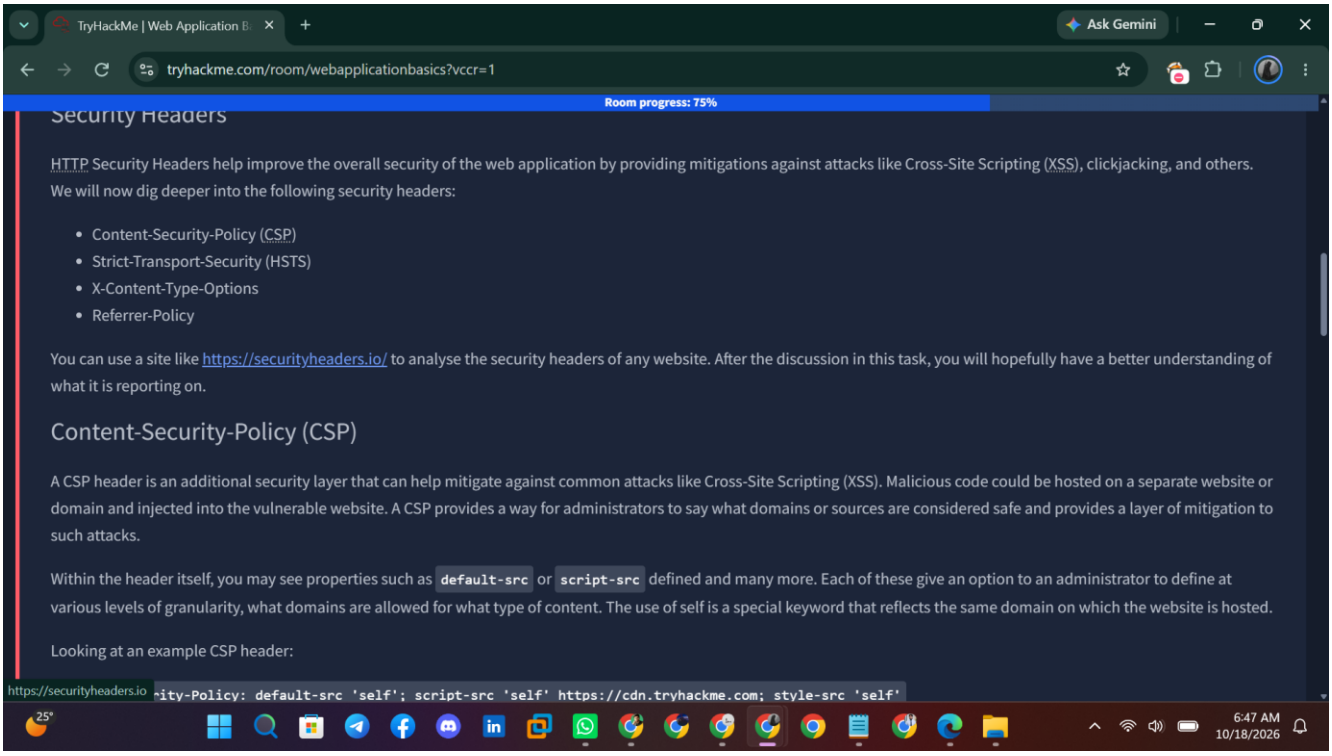


Figure 4: Explore Security headers

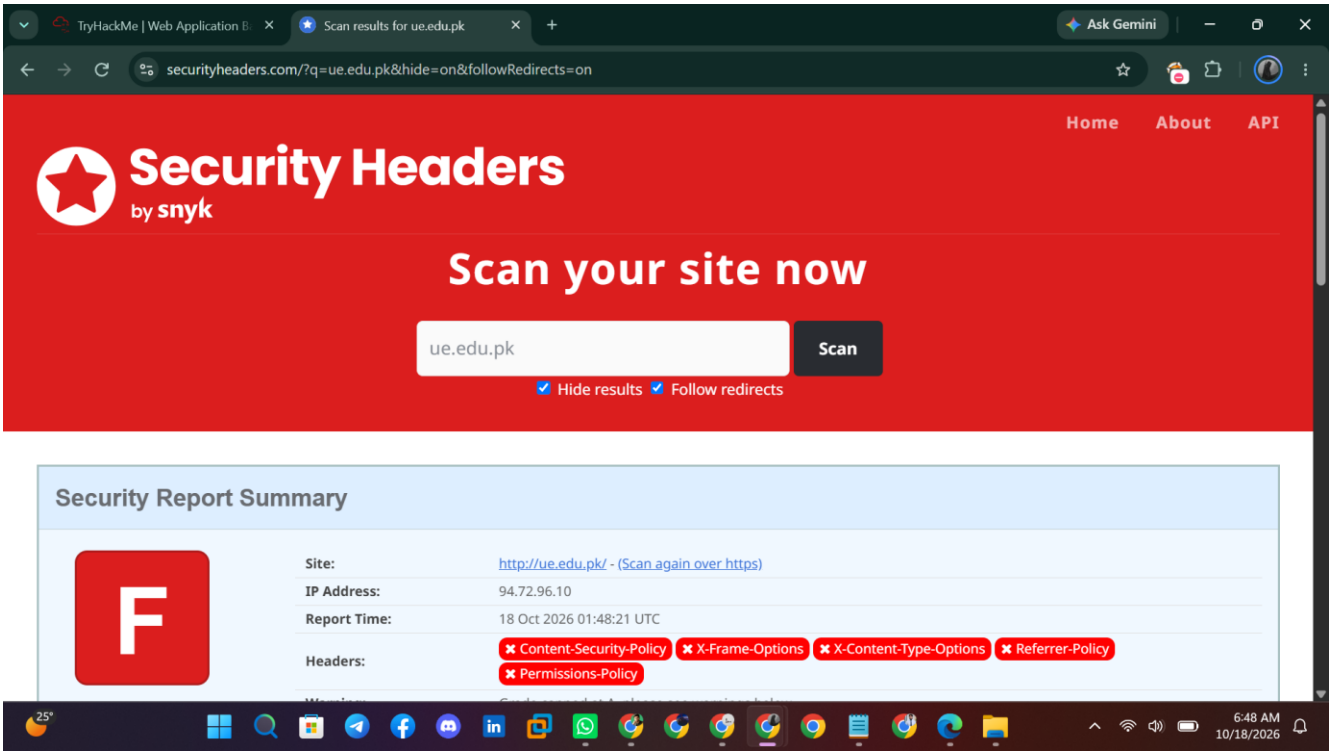


*Figure 5: Scan and find security headers in my own **University website** by using **securityheaders.com***

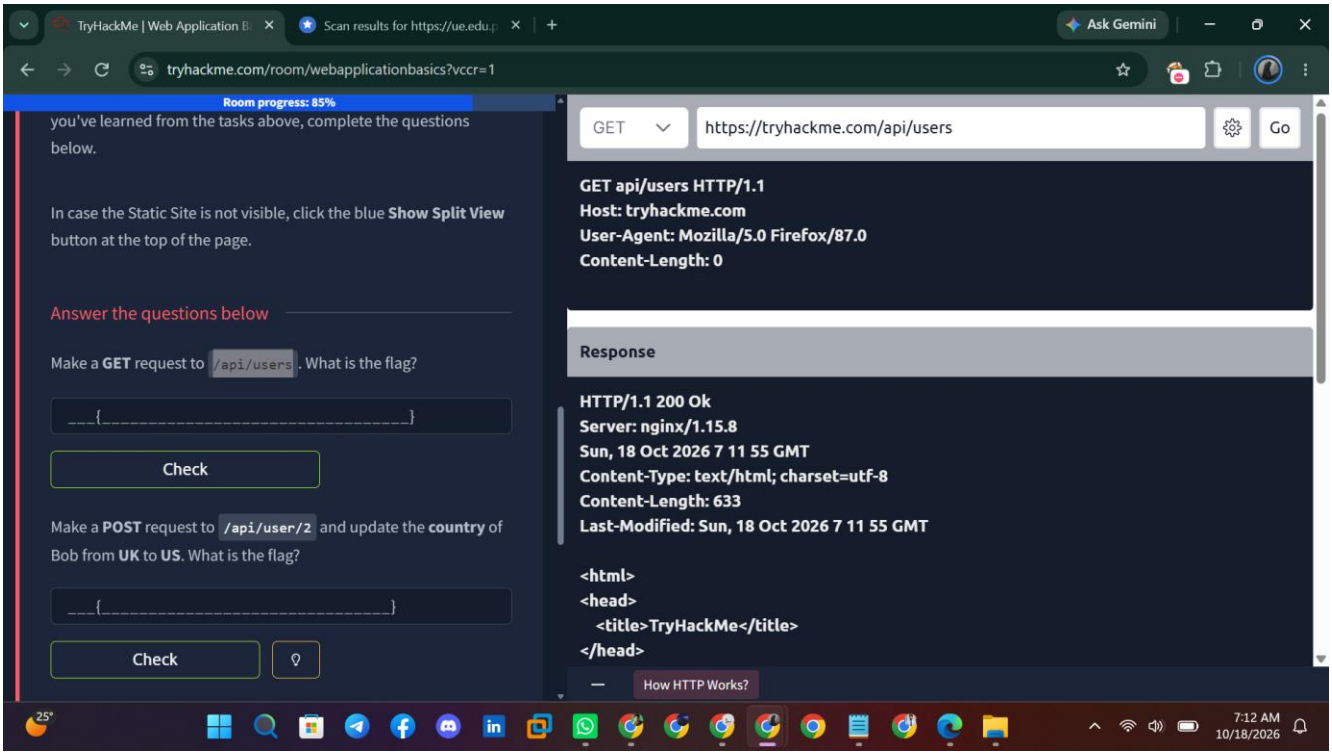


*Figure 6: Make a **GET** request to /api/users*

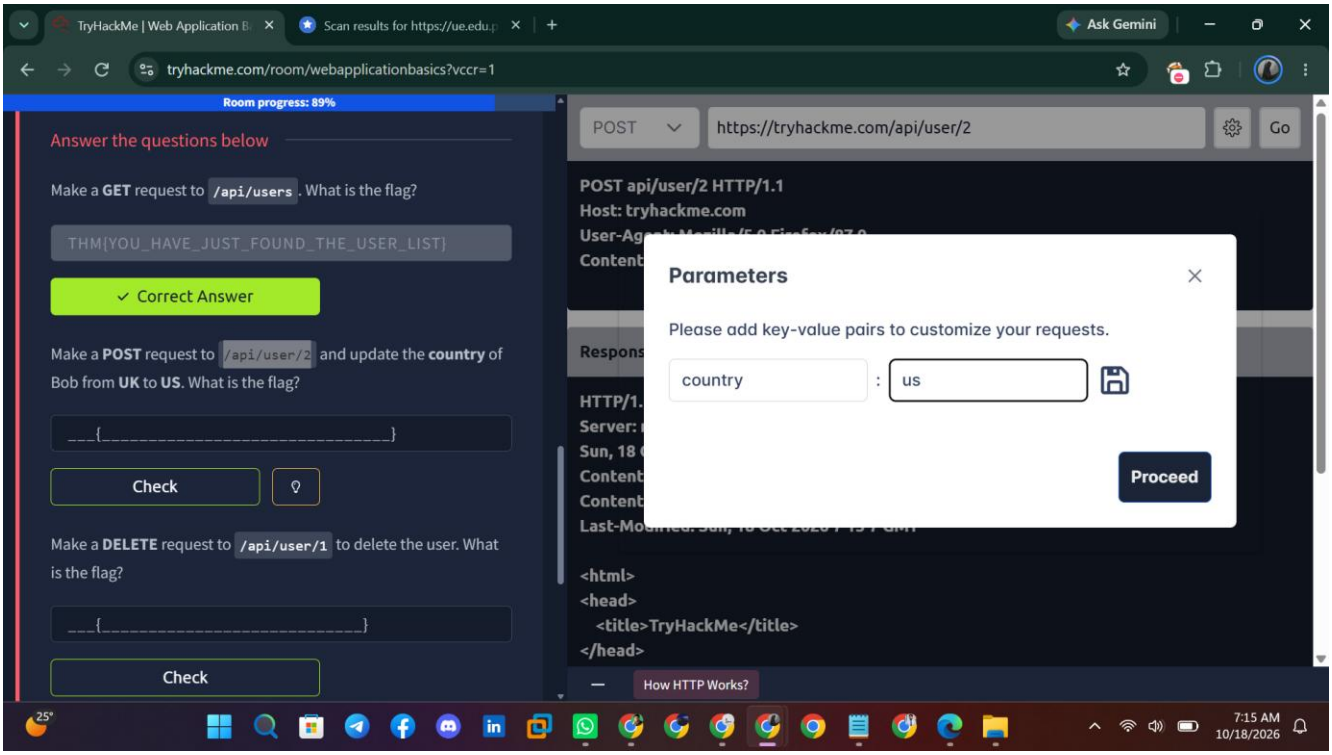


*Figure 7: Make a **POST** request to /api/user/2 and update the country of Bob from UK to US*

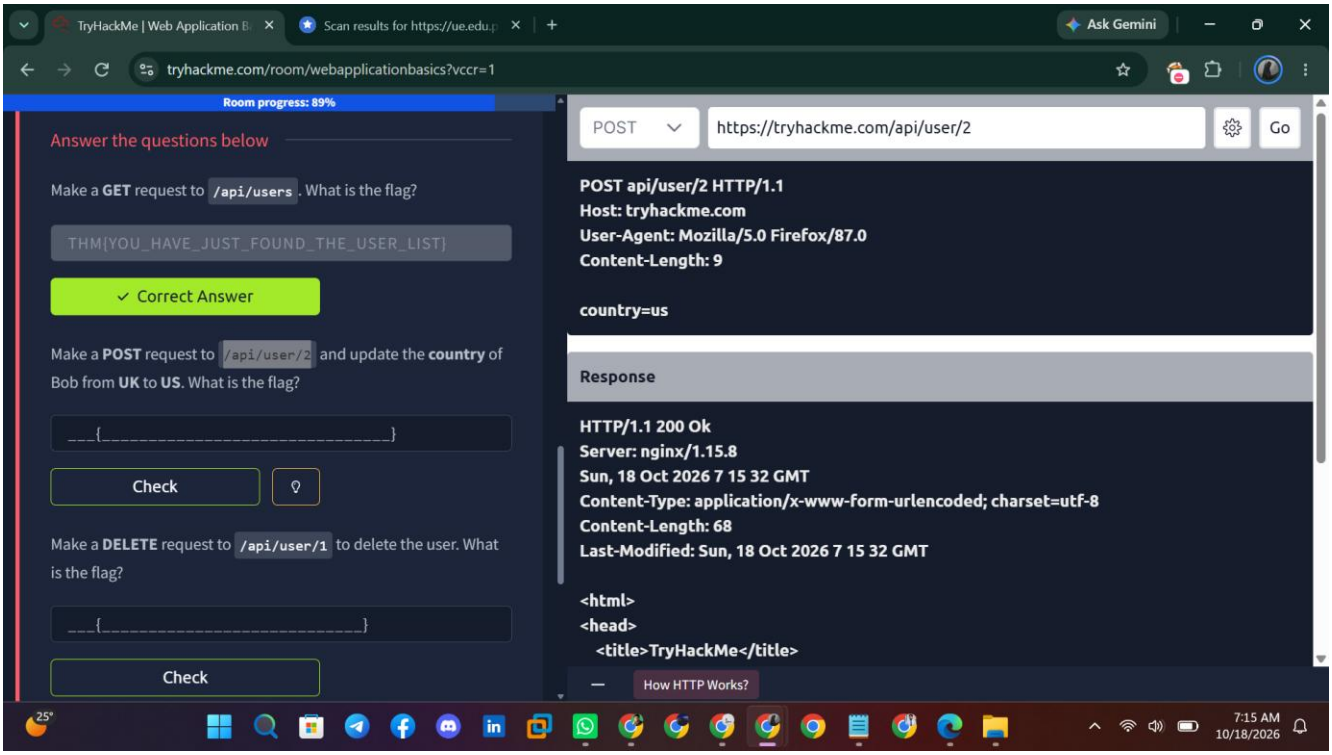


*Figure 8: Watch out **case sensitivity** is matter in some headers*

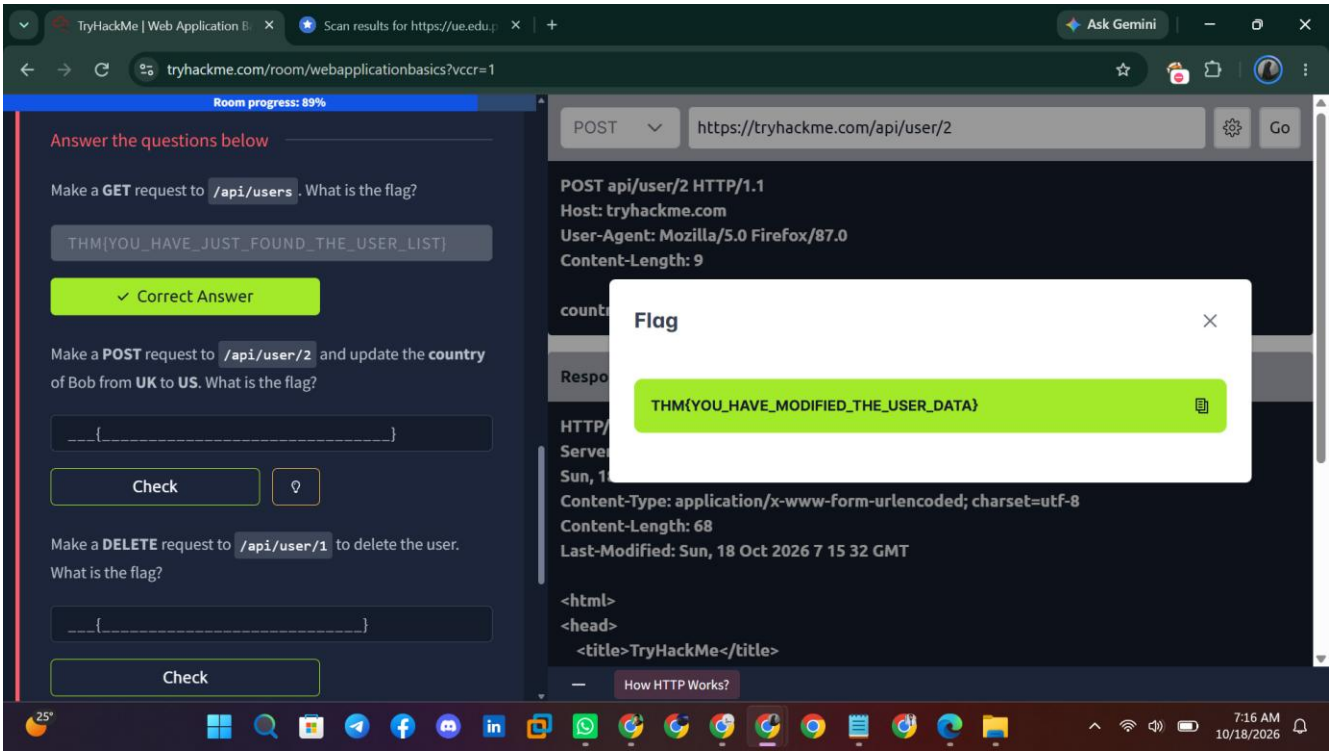


Figure 9: Flag found after use correct case of US

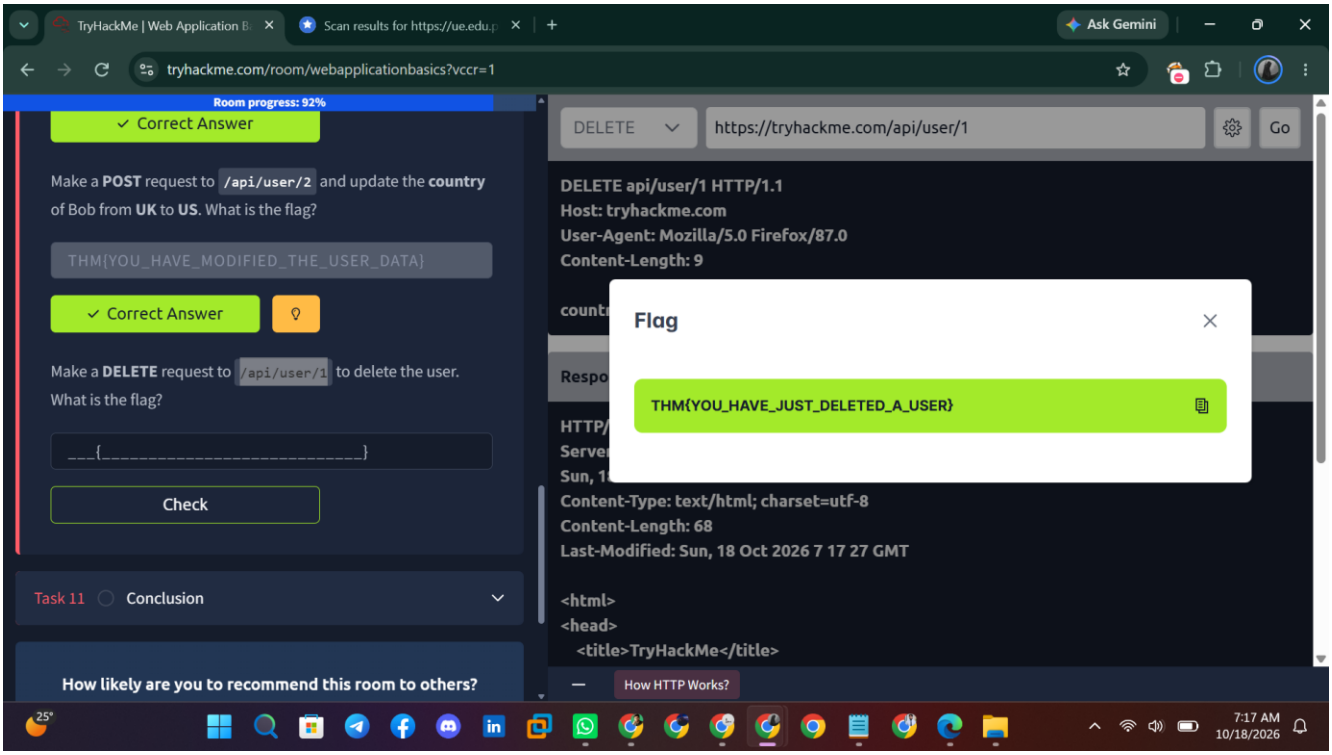


Figure 10: Make a DELETE request to /api/user/1 to delete the user.

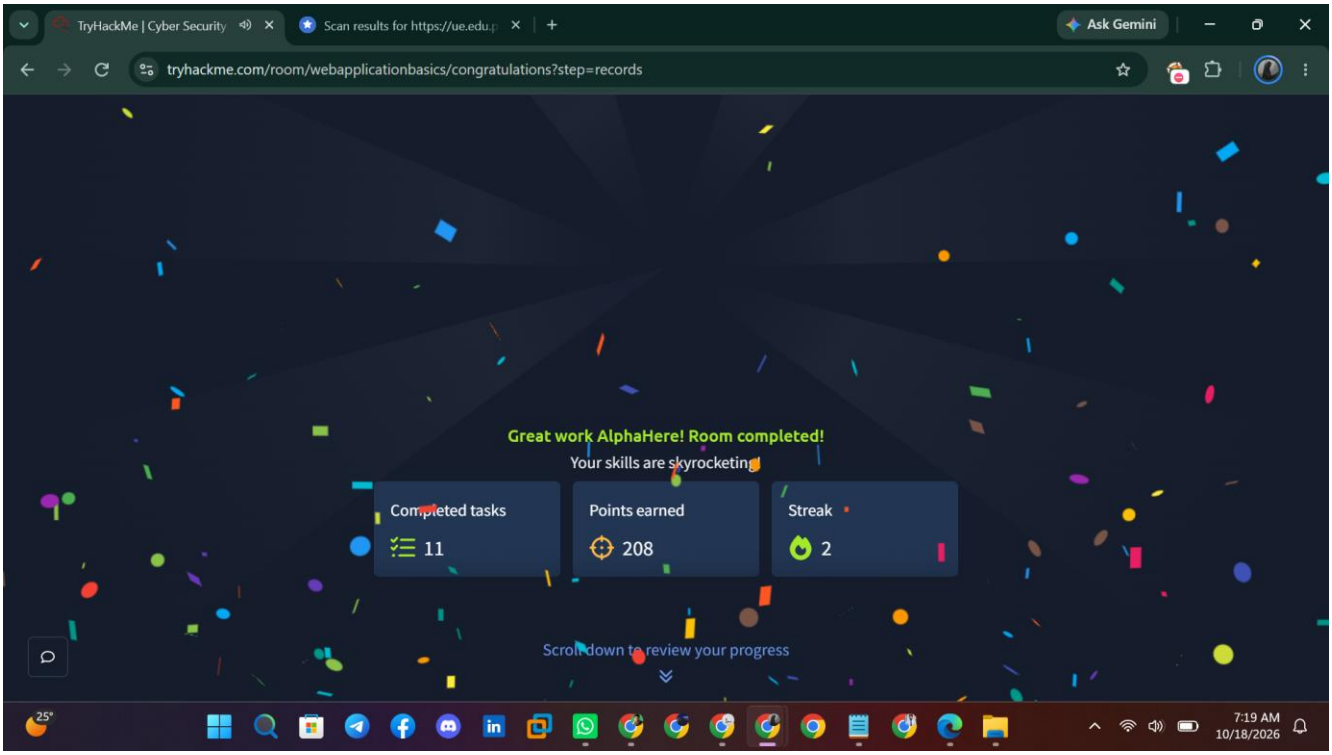


Figure 11: Successfully Solve the room complete all Task and activity of the room

6. Result

I gained practical understanding of HTTP communication and learned how to review security-header information using an online scanning tool. This helped me connect basic web concepts with the initial assessment of web security configurations.

7. Key Learnings

- Understood how browsers and web servers communicate.
- Learned to recognize HTTP methods, status codes, and headers.
- Understood the purpose of security headers and their role in browser protection.
- Gained initial experience reviewing website security-header configurations.

Room 2: HTTP in Detail

1. Objective

To develop a deeper understanding of HTTP and learn how clients and servers exchange information through requests, responses, headers, cookies, and status codes.

2. Concept/Understanding

I learned the structure of HTTP requests and responses, the purpose of common HTTP methods, and how status codes communicate request outcomes. I also explored HTTP headers and cookies to understand how additional information and session-related data are exchanged between clients and servers. I also explore a website in that room that is http.cat that is crazy for understand and memorize the Http codes in easy way.

3. Practical Work

I practiced examining HTTP requests and responses through the room's exercises. I explored different request methods, including GET, POST, PUT, and DELETE, and reviewed examples involving cookies, headers, and HTTP status codes.

4. Tools Used

- TryHackMe
- Web browser
- TryHackMe's provided practice interface
- http.cat
- Inspect tool to analyze cookies

5. Evidence

Relevant screenshots with short captions:

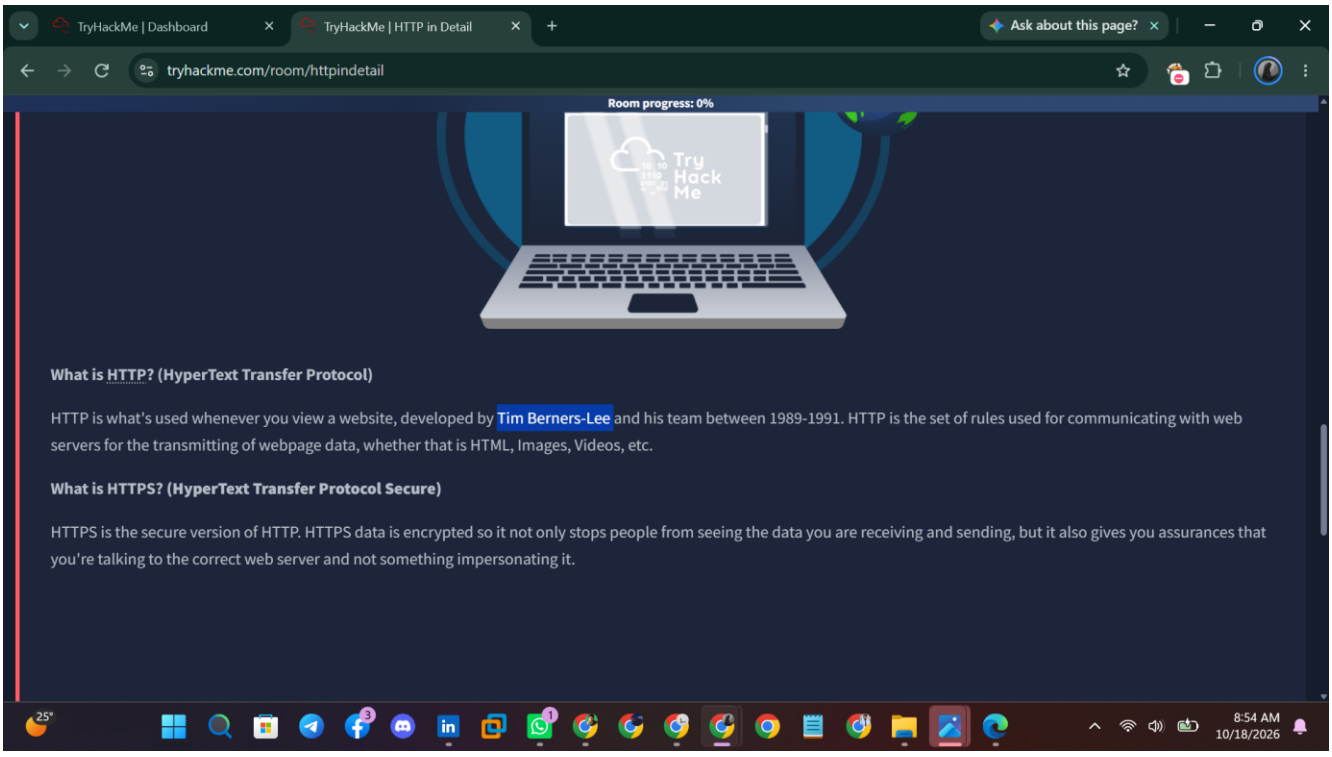


*Figure 1: Http **history***

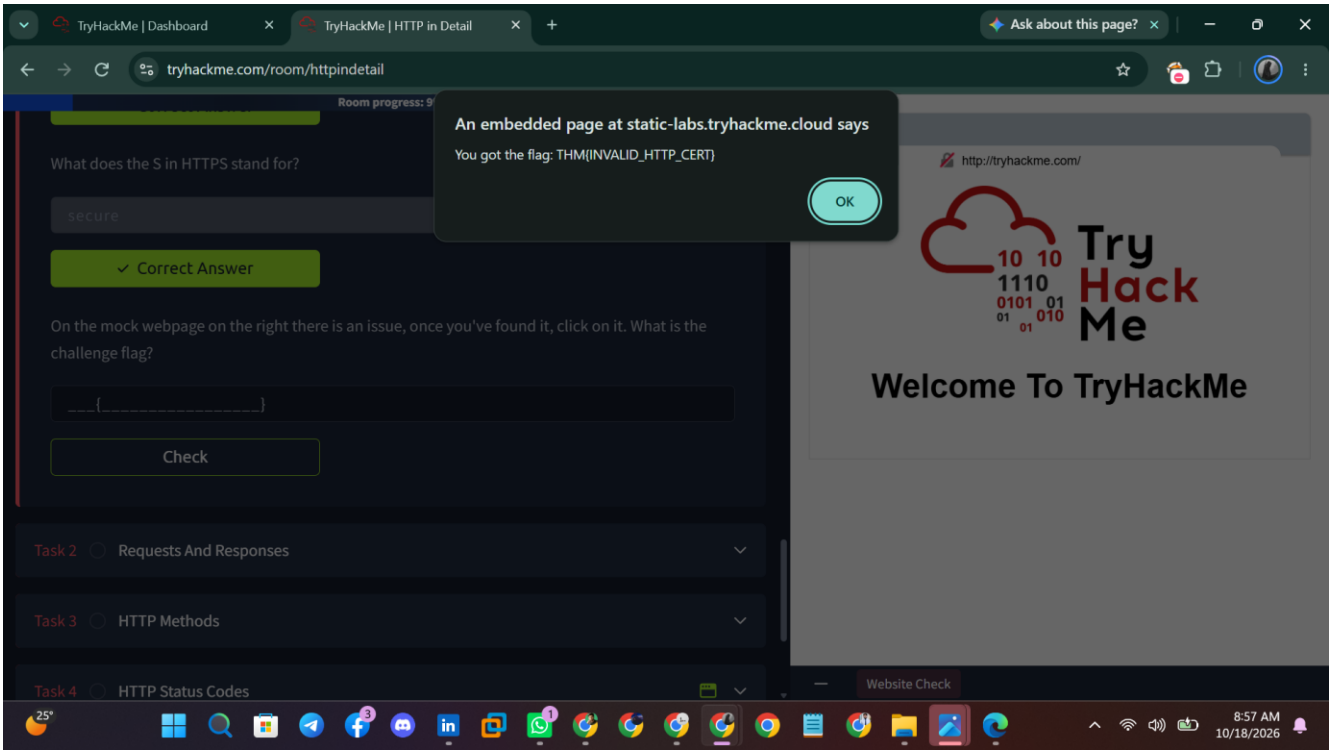


*Figure 2: Intial http **understanding** task*

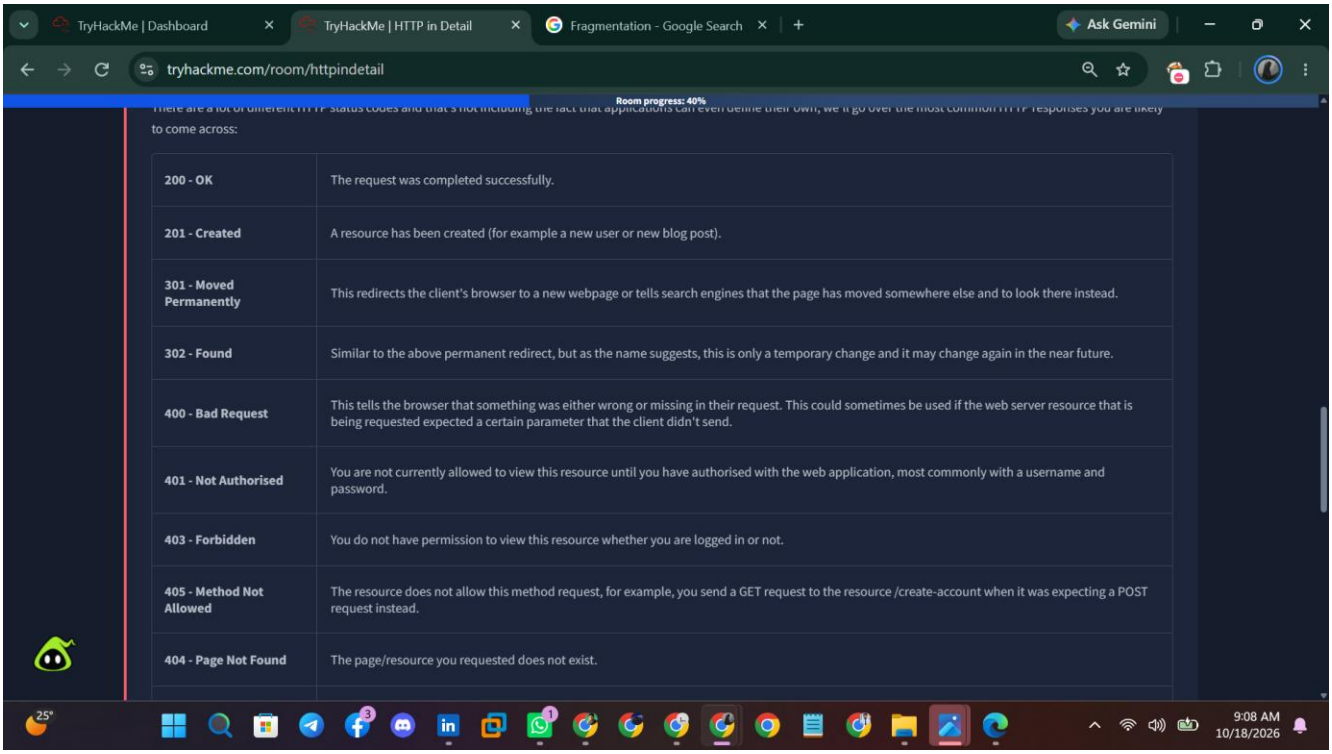


*Figure 3: Common HTTP **codes** or status*

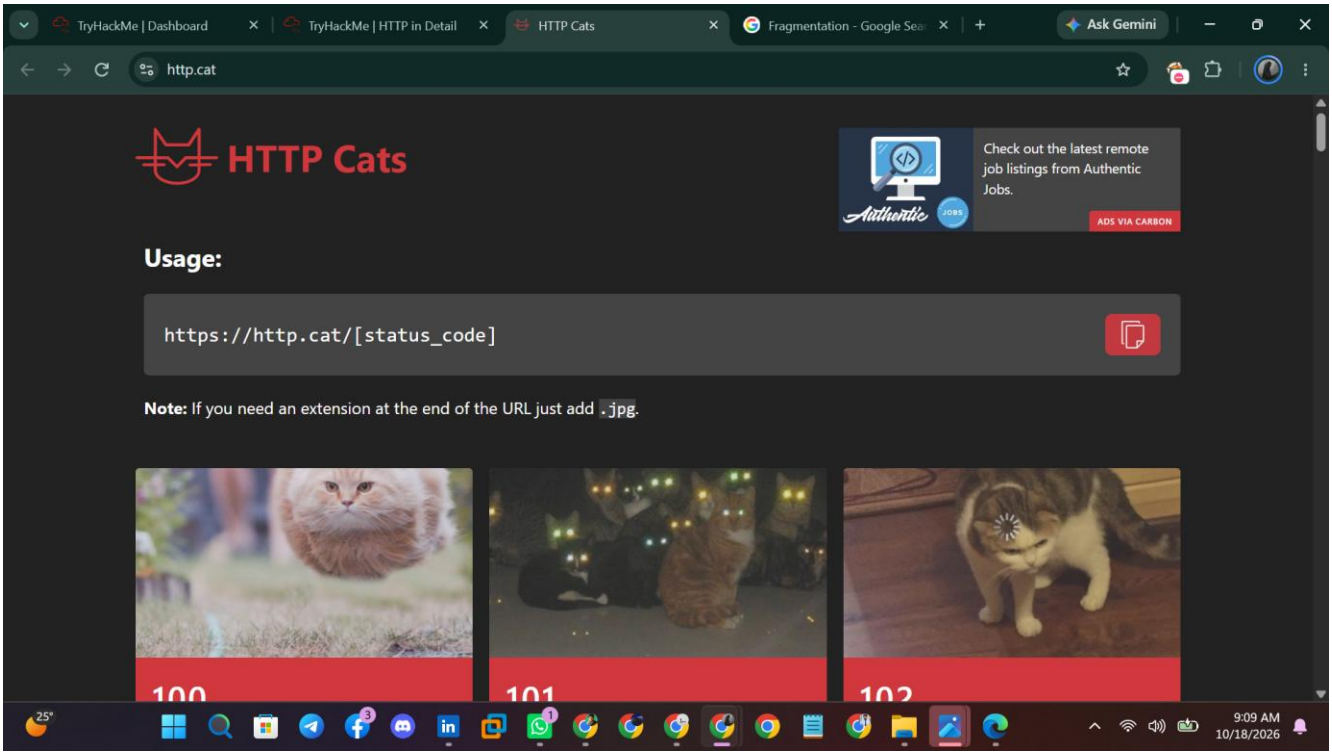


*Figure 4: Explore a website for **understand and memorize http code** in visual method that is **http.cat***

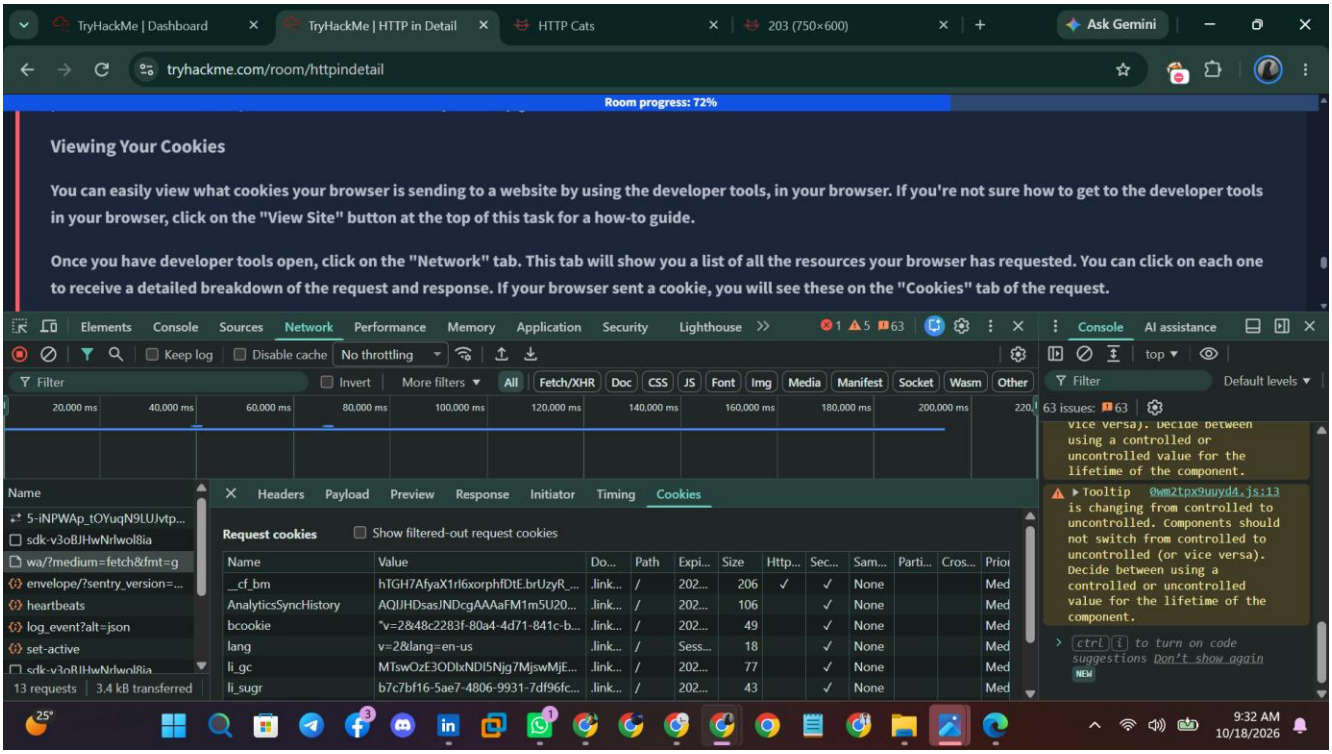


*Figure 5: A small **task** to **understand cookies** in **real time***

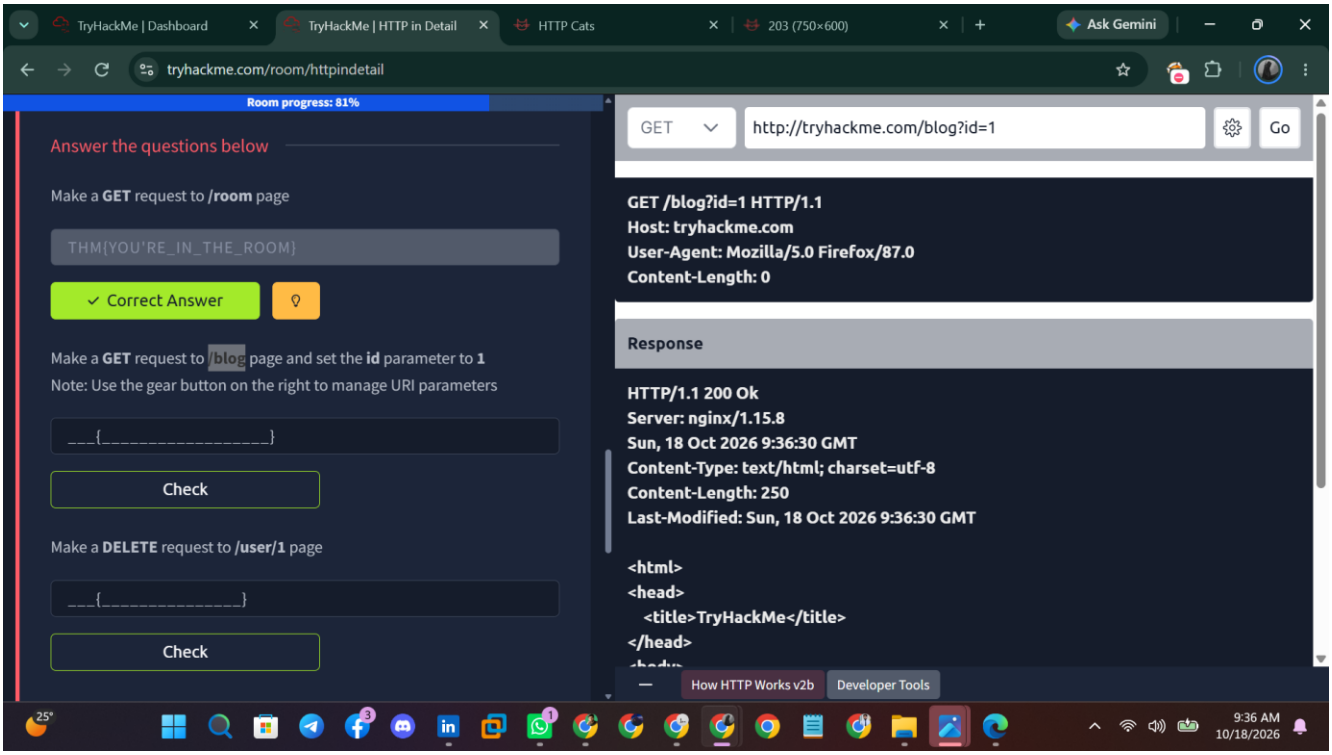


Figure 6: Create a Get Request

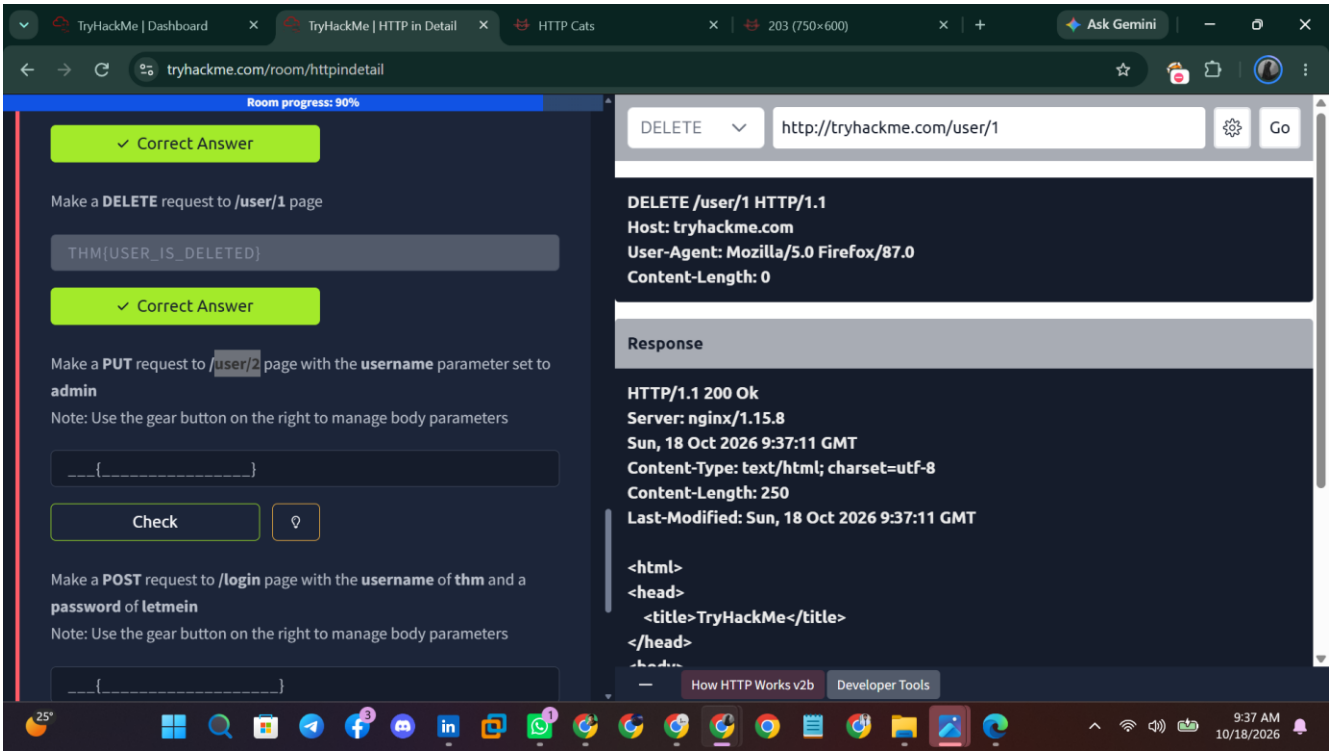


Figure 7: Delete user and understand request http response

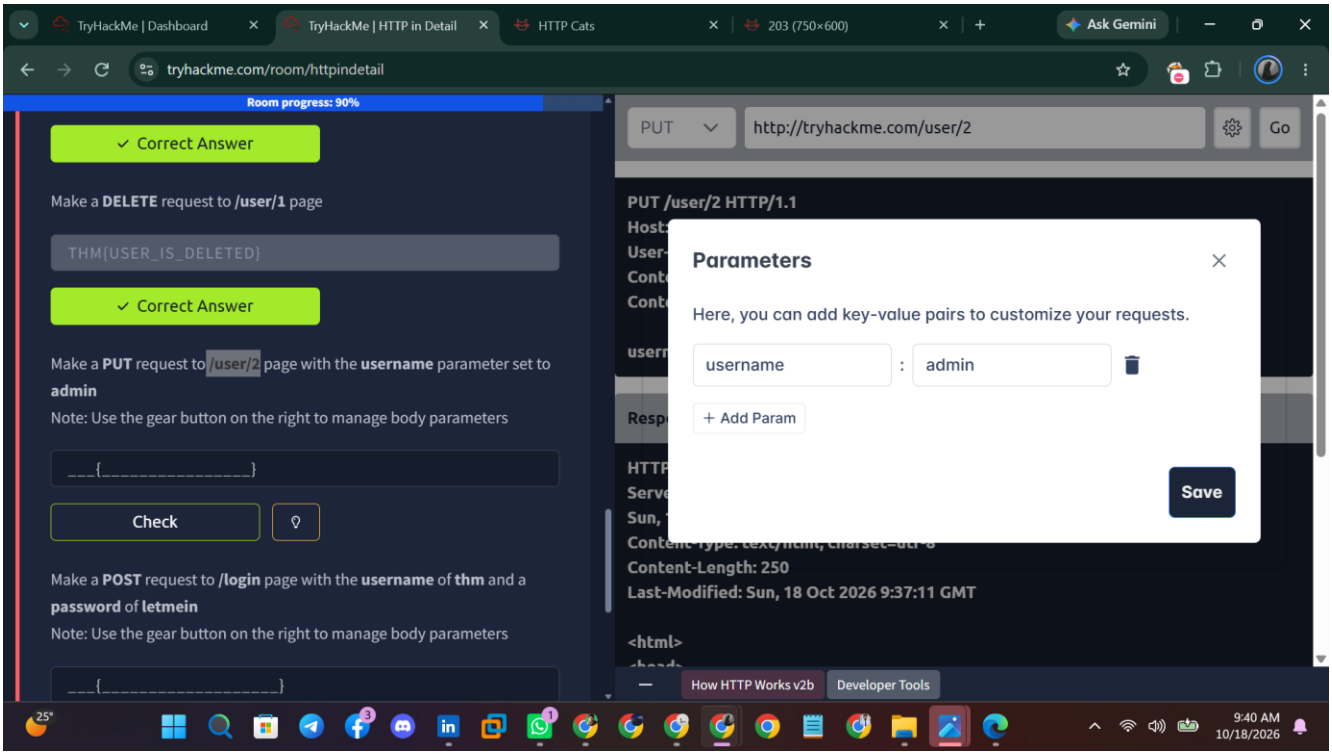


Figure 8: PUT Request

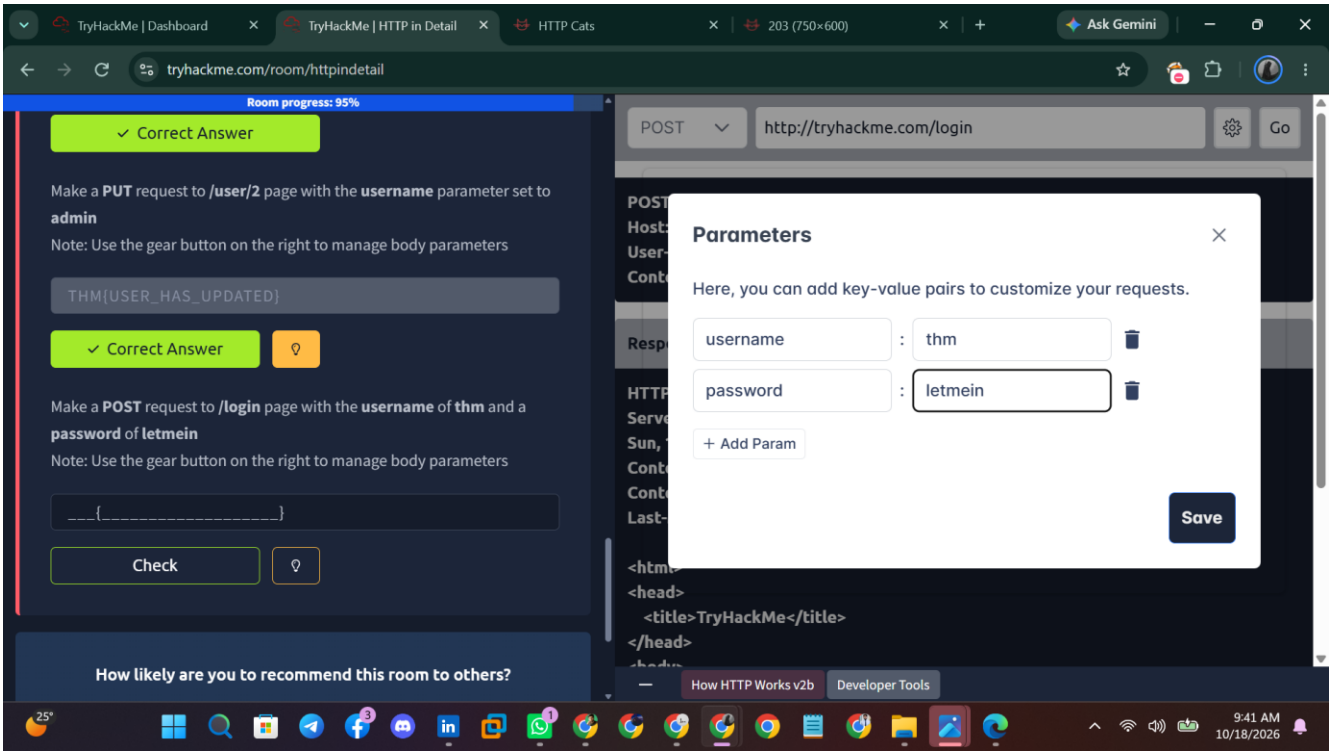


Figure 9: POST HTTP request to login a user with credentials

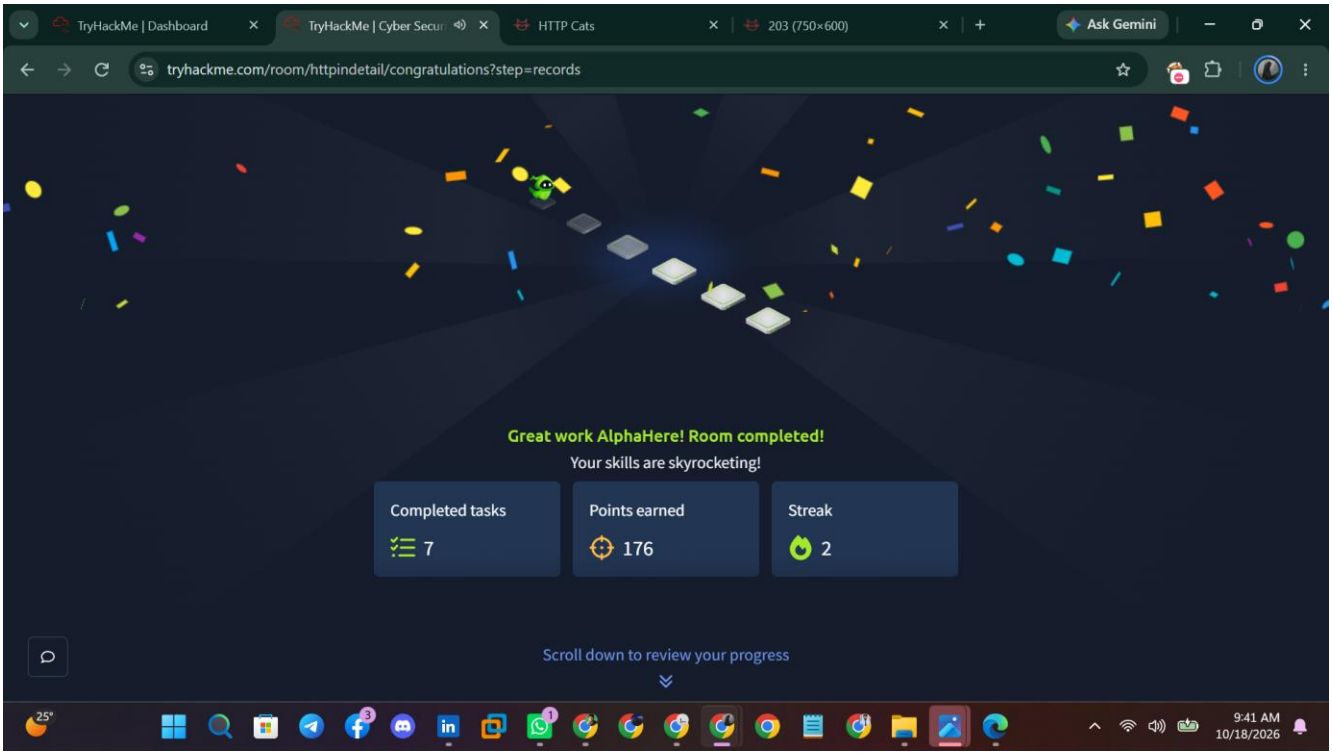


Figure 10 : Compter the Room

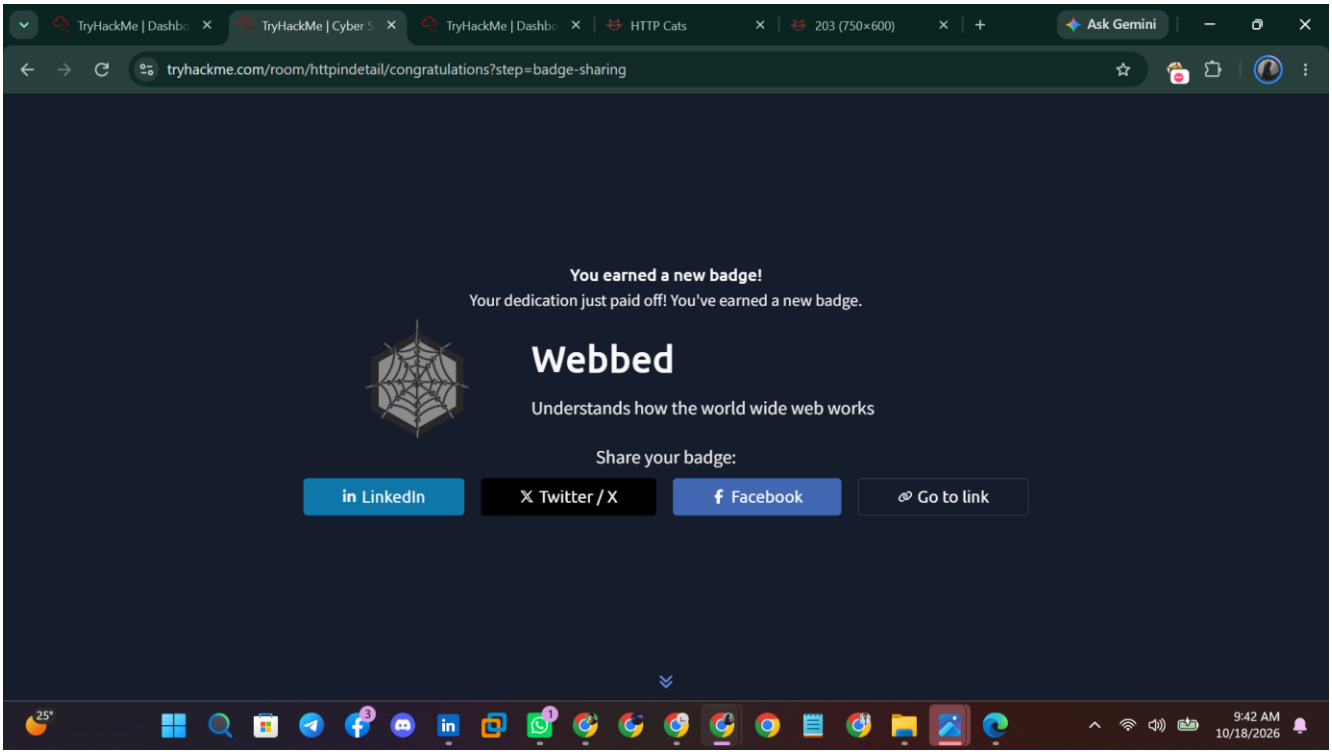


Figure 11: Earn the padge Webbed

6. Result

I improved my understanding of HTTP communication by examining request methods, response codes, headers, and cookies in practical exercises. I also learned how these elements help explain server behavior and application communication.

7. Key Learnings

- Understood the structure of HTTP requests and responses.
- Learned the purposes of common HTTP methods and status codes.
- Improved my understanding of headers and cookies.
- Learned the importance of inspecting server responses when analyzing web application behavior.